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Overview

This document outlines the step-by-step process for the Soil Classification Project. It employs two machine learning approaches: a Convolutional Neural Network (CNN) implemented with TensorFlow, and a Large Language Model (LLM) integrated via Langchain. The results from both methods are compared and contrasted within this document, along with a brief summary of each.

The dataset consists of images obtained from Singapore Site TBM (Tunnel Boring Machine) muck chutes. These images are downloaded from Google Cloud Storage (GCS) and organized by tunnel (Westbound/Eastbound) and ring number. Pre-processing includes cropping and removing duplicate images. Once pre-processed, the data is ready for labeling. All associated scripts are stored in Bitbucket.

Data Pre-Processing

1. Download Dataset

Here is a cleaned dataset of ring P263 – P434 WB.

 [Cleaned Dataset](#)

Images are obtained from GCS. Please request access from your admin.

- If you would like to download all images, please run **gcs_download_all.py**.
- If you would like to download images only from the past 30 days, please run **gcs_download_specific_and_cleaning.py**. Choose the appropriate command to run.

[📄 gcs_download_all.py Documentation](#)

[📄 gcs_download_specific_and_cleaning.py Documentation](#)

2. Clean Dataset

- If you would like to only clean images, please run **gcs_download_specific_and_cleaning.py**. Run commands below:

```
1 find_delete_similar_image(folder)
2
3 for file_name in os.listdir(folder):
4     file_path = os.path.join(folder, file_name)
5     image_preprocessing(file_path, file_name)
```

- If images in your dataset is not in jpg format, you can run **format_converter.py** to convert all image types to jpg.
 - [📄 format_converter.py Documentation](#)
- If you would like to merge the contents of a list of subfolders into a main folder, run the script below.
 - [📄 merge_folder.py Documentation](#)

Machine Learning

Two ML models are developed to test which is better at soil classification.

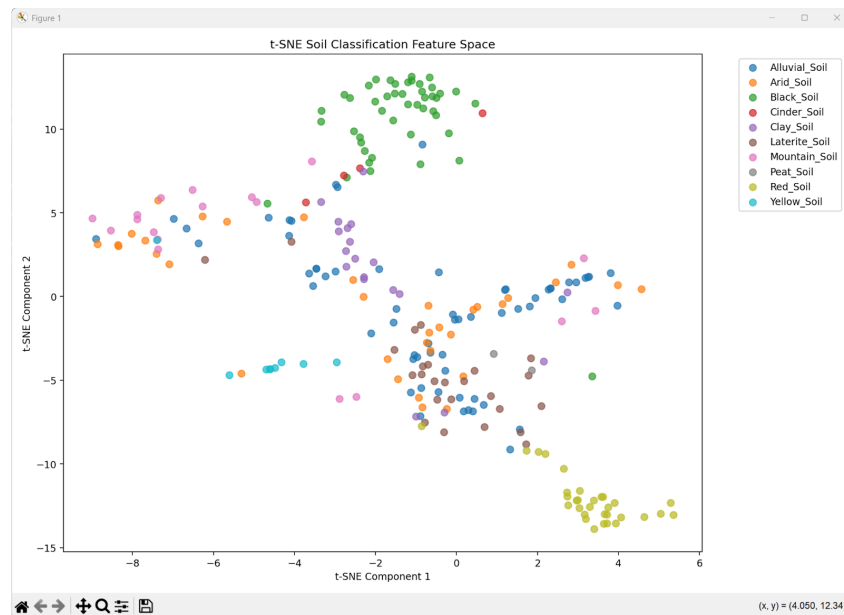
1. Convolutional Neural Network (TensorFlow)

[📄 main_CNN.py Documentation](#)

Accuracy: ~70-80%

Example output: TSNE graph

✓ TSNE for Online Dataset (10 classes) ~70% accuracy



This image displays a t-SNE (t-Distributed Stochastic Neighbour Embedding) plot, titled "t-SNE Soil Classification Feature Space." This type of visualisation is commonly used to reduce the dimensionality of high-dimensional data (like the features extracted by a Convolutional Neural Network, or CNN) into a 2D or 3D space, allowing for visual inspection of how well different classes are separated.

Overall Purpose: The plot aims to visualise the feature representations learned by a CNN for different soil types. Each point on the graph represents a soil image, and its position is determined by the similarity of its learned features to other samples. **Points that are close together in this 2D space have similar features, while points far apart have dissimilar features. The goal for a good classification model is to have distinct, well-separated clusters for each class.**

Key Observations and Analysis:

1. General Separability:

- **Some classes are very well separated:** The plot shows several distinct clusters, suggesting that the CNN has learned highly discriminative features for certain soil types.
- **Other classes show significant overlap:** There are also areas where points from different soil types are intermingled, indicating that the CNN struggles to differentiate between these specific classes based on their learned features.

2. Well-Separated Clusters (Good Performance):

- **Peat_Soil (Green):** This class forms a very tight and distinct cluster in the upper-middle part of the graph (around t-SNE Component 1 = 0 to 2, t-SNE Component 2 =

8 to 12). This suggests the CNN is highly effective at identifying and distinguishing Peat Soil from all other types.

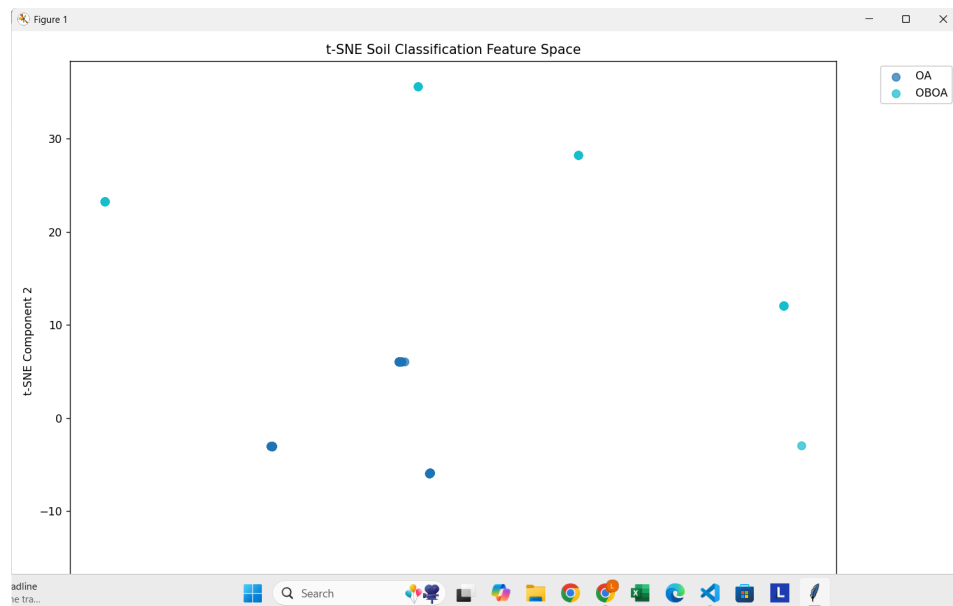
- **Yellow_Soil (Light Yellow/Gold):** This class also forms a very compact and isolated cluster in the bottom-right corner (around t-SNE Component 1 = 3 to 5, t-SNE Component 2 = -10 to -15). This indicates excellent separability for Yellow Soil.
- **Alluvial_Soil (Blue) and Arid_Soil (Orange) (Left Side):** On the far left side of the plot (t-SNE Component 1 = -9 to -7), Alluvial and Arid soils form relatively distinct, though somewhat elongated, clusters. While there's some minor intermingling, they are generally well-separated from most other classes.

3. **Overlapping/Intermingled Clusters (Potential Challenges):**

- **Central Cluster:** A large, dense cluster in the centre of the plot (roughly t-SNE Component 1 = -2 to 2, t-SNE Component 2 = -5 to 5) contains a mix of several soil types, including:
 - Clay_Soil (Purple)
 - Laterite_Soil (Brown)
 - Red_Soil (Red)
 - Mountain_Soil (Grey)
 - Cinder_Soil (Pink)
 - Some Alluvial_Soil (Blue) and Arid_Soil (Orange)
- **Interpretation of Overlap:** The significant overlap among these classes suggests that the CNN's learned features for these soil types are not sufficiently distinct. This could lead to misclassifications between these particular soil types. For example, it might be difficult for the CNN to reliably distinguish between Clay, Laterite, and Red soils based on the features it has learned.

4. **Outliers/Scattered Points:** Some individual points from otherwise well-clustered classes might appear far from their main cluster (e.g., a few green Peat_Soil points scattered among other clusters). These could represent:

- Misabeled data in the training set.
- Unusual examples within that class that the CNN struggles to categorise correctly.
- Samples that genuinely share features with other classes.



Key Observations:

1. Clear Class Separation:

- The plot shows good separation between the two soil classes
- OA samples (dark blue) are predominantly clustered in the lower portion of the plot (t-SNE Component 2 values around -5 to -10)
- OBOA samples (light blue/cyan) are distributed in the upper and right portions of the plot (t-SNE Component 2 values around 5 to 35)

2. Cluster Characteristics:

- **OA Class:** Forms a relatively tight, compact cluster in the lower-left quadrant, suggesting consistent feature representations
- **OBOA Class:** Shows more dispersed distribution but still maintains clear separation from OA, with samples spread across the upper portion of the feature space

3. Minimal Overlap:

- There is virtually no overlap between the two classes in this 2D representation
- This indicates the CNN has learned highly discriminative features that can effectively distinguish between OA and OBOA soil types (as a result of 20 epochs).

Implications for CNN Performance:

This t-SNE visualization provides strong evidence that the **CNN is effective** for this specific OA vs. OBOA classification task:

- **High Expected Accuracy:** The clear separation suggests the CNN should achieve very high accuracy (likely >95%) on this binary classification

- **Robust Feature Learning:** The distinct clusters indicate the CNN has learned meaningful, discriminative visual features
- **Low Misclassification Risk:** The minimal overlap suggests very few samples would be misclassified

Justification for CNN Superiority Using Accuracy:

Based on this t-SNE analysis, here's why the **CNN is superior to the LLM** for achieving high classification accuracy:

1. **Demonstrated Feature Separability:**

- The t-SNE plot provides visual proof that the CNN has learned features that create clear, well-separated clusters for OA and OBOA
- This level of separation directly translates to high classification accuracy because the decision boundary between classes is clear and robust

2. **Robust Decision Boundaries:**

- The clear separation suggests the CNN has learned **stable, generalisable features** that should maintain high accuracy on new, unseen soil samples
- The LLM's variable confidence scores (1/10 to 9/10) for the same class suggest less consistent internal representations

In contrast, while the LLM provided rich descriptions, its highly variable confidence scores (ranging from 1/10 to 9/10) even for correct classifications suggest less reliable internal feature representations compared to the CNN's clearly separated feature clusters.

2. Large Language Model (Langchain)

 [main_LLM.py Documentation](#)

Accuracy: ~15%

Example output: Terminal Print

✓ Print result for OA

processing C:\Users\Tunnel\Documents\Soil_Identification\GCS Training\OA...

[file: processed_CREC910WB_R396_1395mm_C1_2025-06-24_04-35.jpg](#)

name='1' colour='Light grey to off-white, with some darker grey areas from shadows.'

wetness='Dry and powdery.' size='Fine to coarse sand, with no distinct gravels visible in this view.' confidence='6/10'

[file: processed_CREC910WB_R401_1395mm_C1_2025-06-24 18-49.jpg](#)

name='2' colour='Light grey' wetness='Damp' size='Fine-grained, cohesive'
confidence='3/10'

[file: processed_CREC910WB_R402_995mm_C1_2025-06-24 21-09.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Damp' size='Fine to coarse sand with fine gravels' confidence='8/10'

[file: processed_CREC910WB_R403_1295mm_C1_2025-06-24 23-03.jpg](#)

name='2' colour='Greenish grey' wetness='Damp' size='Fine to coarse sand, very clayey'
confidence='8/10'

[file: processed_CREC910WB_R406_1495mm_C1_2025-06-25 07-17.jpg](#)

name='2' colour='Predominantly greyish, with subtle greenish-brown undertones.'
wetness='Damp' size='Predominantly fine-grained, appearing very clayey, with no clearly discernible coarse sand or gravel.' confidence='8/10'

[file: processed_CREC910WB_R406_1895mm_C1_2025-06-25 07-27.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Damp' size='Fine to coarse sand' confidence='8/10'

[file: processed_CREC910WB_R408_1695mm_C1_2025-06-25 23-06.jpg](#)

name='2' colour='Greenish grey' wetness='Dry' size='Fine to coarse sand'
confidence='8/10'

[file: processed_CREC910WB_R409_1195mm_C1_2025-06-26 00-31.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Damp' size='Fine to coarse sand' confidence='9'

[file: processed_CREC910WB_R409_1595mm_C1_2025-06-26 00-41.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Wet' size='Fine to coarse sand with fine gravels' confidence='8'

[file: processed_CREC910WB_R411_895mm_C1_2025-06-26 05-15.jpg](#)

name='2' colour='Greenish-grey and yellowish-brown' wetness='Very wet, like a slurry or mudflow' size='Fine to coarse sand with fine gravels, very clayey' confidence='8/10'

[file: processed_CREC910WB_R412_1595mm_C1_2025-06-26 12-27.jpg](#)

name='2' colour='greenish grey and yellowish brown' wetness='Damp and cohesive' size='Fine to coarse sand, with fine gravels' confidence='8'

[file: processed_CREC910WB_R415_1695mm_C1_2025-06-27 01-14.jpg](#)

name='1' colour='yellowish brown mottled with light greenish grey' wetness='dry to slightly damp' size='medium to coarse sand' confidence='8/10'

[file: processed_CREC910WB_R433_1895mm_C1_2025-06-30 10-33.jpg](#)

name='2' colour='Greenish grey and yellowish brown (appears predominantly greyish in the image)' wetness='Very wet / Submerged' size='Fine to coarse sand with very clayey matrix and slight gravel' confidence='8'

✓ Print result for OBOA

processing C:\Users\Tunnel\Documents\Soil_Identification\GCS Training\OBOA...

[file: processed_CREC910WB_R365_1695mm_C1_2025-06-17 13-01.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Very wet, high moisture content (slurry-like)' size='Fine to coarse sand' confidence='9/10'

[file: processed_CREC910WB_R366_1695mm_C1_2025-06-17](#) 15-28.jpg

name='2' colour='Greenish grey and yellowish brown' wetness='Wet' size='Fine to coarse sand with slight gravels' confidence='8/10'

[file: processed_CREC910WB_R376_995mm_C1_2025-06-18](#) 17-34.jpg

name='2' colour='Light grey to white' wetness='Very dry and powdery' size='Very fine-grained (appears like silt or very fine sand)' confidence='2/10'

[file: processed_CREC910WB_R381_895mm_C1_2025-06-19](#) 11-14.jpg

name='2' colour='Light grey' wetness='Dry' size='Very fine, powdery (no visible sand or gravel)' confidence='3'

[file: processed_CREC910WB_R383_1495mm_C1_2025-06-19](#) 16-24.jpg

name='2' colour='Light brownish-grey' wetness='Dry' size='Fine to coarse sand, with significant clay content' confidence='6/10'

[file: processed_CREC910WB_R387_1895mm_C1_2025-06-20](#) 04-57.jpg

name='2' colour='Light grey, with subtle yellowish-brown and greenish-grey tones.' wetness='Very dry, powdery.' size='Fine-grained, appearing as very fine sand or clay.' confidence='6/10'

[file: processed_CREC910WB_R387_995mm_C1_2025-06-20](#) 04-34.jpg

name='2' colour='greenish grey and yellowish brown' wetness='very wet' size='fine to coarse sand' confidence='8'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20](#) 06-58.jpg

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20 07-03.jpg](#)

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

processing C:\Users\Tunnel\Documents\Soil_Identification\GCS Training\OBOA...

[file: processed_CREC910WB_R365_1695mm_C1_2025-06-17 13-01.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Very wet, high moisture content (slurry-like)' size='Fine to coarse sand' confidence='9/10'

[file: processed_CREC910WB_R366_1695mm_C1_2025-06-17 15-28.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Wet' size='Fine to coarse sand with slight gravels' confidence='8/10'

[file: processed_CREC910WB_R376_995mm_C1_2025-06-18 17-34.jpg](#)

name='2' colour='Light grey to white' wetness='Very dry and powdery' size='Very fine-grained (appears like silt or very fine sand)' confidence='2/10'

[file: processed_CREC910WB_R381_895mm_C1_2025-06-19 11-14.jpg](#)

name='2' colour='Light grey' wetness='Dry' size='Very fine, powdery (no visible sand or gravel)' confidence='3'

[file: processed_CREC910WB_R383_1495mm_C1_2025-06-19 16-24.jpg](#)

name='2' colour='Light brownish-grey' wetness='Dry' size='Fine to coarse sand, with significant clay content' confidence='6/10'

[file: processed_CREC910WB_R387_1895mm_C1_2025-06-20 04-57.jpg](#)

name='2' colour='Light grey, with subtle yellowish-brown and greenish-grey tones.' wetness='Very dry, powdery.' size='Fine-grained, appearing as very fine sand or clay.'

confidence='6/10'

[file: processed_CREC910WB_R387_995mm_C1_2025-06-20 04-34.jpg](#)

name='2' colour='greenish grey and yellowish brown' wetness='very wet' size='fine to coarse sand' confidence='8'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20 06-58.jpg](#)

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20 07-03.jpg](#)

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

[file: processed_CREC910WB_R366_1695mm_C1_2025-06-17 15-28.jpg](#)

name='2' colour='Greenish grey and yellowish brown' wetness='Wet' size='Fine to coarse sand with slight gravels' confidence='8/10'

[file: processed_CREC910WB_R376_995mm_C1_2025-06-18 17-34.jpg](#)

name='2' colour='Light grey to white' wetness='Very dry and powdery' size='Very fine-grained (appears like silt or very fine sand)' confidence='2/10'

[file: processed_CREC910WB_R381_895mm_C1_2025-06-19 11-14.jpg](#)

name='2' colour='Light grey' wetness='Dry' size='Very fine, powdery (no visible sand or gravel)' confidence='3'

[file: processed_CREC910WB_R383_1495mm_C1_2025-06-19 16-24.jpg](#)

name='2' colour='Light brownish-grey' wetness='Dry' size='Fine to coarse sand, with significant clay content' confidence='6/10'

[file: processed_CREC910WB_R387_1895mm_C1_2025-06-20 04-57.jpg](#)

name='2' colour='Light grey, with subtle yellowish-brown and greenish-grey tones.' wetness='Very dry, powdery.' size='Fine-grained, appearing as very fine sand or clay.' confidence='6/10'

[file: processed_CREC910WB_R387_995mm_C1_2025-06-20 04-34.jpg](#)

name='2' colour='greenish grey and yellowish brown' wetness='very wet' size='fine to coarse sand' confidence='8'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20 06-58.jpg](#)

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20 07-03.jpg](#)

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

[file: processed_CREC910WB_R381_895mm_C1_2025-06-19 11-14.jpg](#)

name='2' colour='Light grey' wetness='Dry' size='Very fine, powdery (no visible sand or gravel)' confidence='3'

[file: processed_CREC910WB_R383_1495mm_C1_2025-06-19 16-24.jpg](#)

name='2' colour='Light brownish-grey' wetness='Dry' size='Fine to coarse sand, with significant clay content' confidence='6/10'

[file: processed_CREC910WB_R387_1895mm_C1_2025-06-20 04-57.jpg](#)

name='2' colour='Light grey, with subtle yellowish-brown and greenish-grey tones.' wetness='Very dry, powdery.' size='Fine-grained, appearing as very fine sand or clay.' confidence='6/10'

[file: processed_CREC910WB_R387_995mm_C1_2025-06-20 04-34.jpg](#)

name='2' colour='greenish grey and yellowish brown' wetness='very wet' size='fine to coarse sand' confidence='8'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20 06-58.jpg](#)

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20 07-03.jpg](#)

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

name='2' colour='Light brownish-grey' wetness='Dry' size='Fine to coarse sand, with significant clay content' confidence='6/10'

[file: processed_CREC910WB_R387_1895mm_C1_2025-06-20 04-57.jpg](#)

name='2' colour='Light grey, with subtle yellowish-brown and greenish-grey tones.' wetness='Very dry, powdery.' size='Fine-grained, appearing as very fine sand or clay.' confidence='6/10'

[file: processed_CREC910WB_R387_995mm_C1_2025-06-20 04-34.jpg](#)

name='2' colour='greenish grey and yellowish brown' wetness='very wet' size='fine to coarse sand' confidence='8'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20 06-58.jpg](#)

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20 07-03.jpg](#)

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

[file: processed_CREC910WB_R387_995mm_C1_2025-06-20 04-34.jpg](#)

name='2' colour='greenish grey and yellowish brown' wetness='very wet' size='fine to coarse sand' confidence='8'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20 06-58.jpg](#)

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20 07-03.jpg](#)

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20](#) 06-58.jpg

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20](#) 07-03.jpg

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

[file: processed_CREC910WB_R388_795mm_C1_2025-06-20](#) 06-58.jpg

name='2' colour='Light grey' wetness='Dry, powdery' size='Very fine-grained, appears like powder, no visible sand or gravel' confidence='1/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20](#) 07-03.jpg

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

[file: processed_CREC910WB_R388_995mm_C1_2025-06-20](#) 07-03.jpg

name='2' colour='Predominantly light grey, consistent with a greenish-grey tone, with darker, wet patches.' wetness='Visibly wet in some areas, particularly on the left where water appears to be flowing or pooling, while the bulk of the soil appears damp and cohesive.' size='Predominantly fine-grained, appearing very clayey, consistent with the description of OBOA.' confidence='8/10'

[file: processed_CREC910WB_R393_1695mm_C1_2025-06-23](#) 20-44.jpg

name='2' colour='Light greyish-brown' wetness='Dry' size='Predominantly fine-grained, appearing clumpy, consistent with a very clayey soil.' confidence='8/10'

name='2' colour='Greenish grey and yellowish brown' wetness='Wet, with some standing water visible' size='Fine to coarse sand with fine gravels' cofidence='8/10'

Key Observations and Analysis:

1. Varying Confidence Levels:

- While the classification is consistently '2', the **confidence** scores vary significantly:
 - High confidence: 9/10 , 8/10 , 8
 - Medium confidence: 6/10
 - Low confidence: 2/10 , 3 , 1/10
- **Interpretation:** This variation is crucial. Even when the model correctly identifies the soil as OBOA, its certainty is not constant. Low confidence scores (e.g., 1/10 , 2/10 , 3) indicate that for those specific images, the model found it challenging to make a confident prediction, despite arriving at the correct class. This could be due to:
 - Ambiguous visual features in the image.
 - The image representing a less typical example of OBOA.
 - Limitations in the model's ability to discern subtle differences for certain soil conditions (e.g., very dry/powdery vs. wet/slurry-like).

2. Rich Descriptive Feature Extraction:

- The model successfully extracts detailed qualitative descriptions for **colour** , **wetness** , and **size** for each image. These descriptions are varied and appear to be relevant to soil characteristics:
 - **Colour:** Ranges from "Greenish grey and yellowish brown" to "Light grey to white" and "Light brownish-grey."
 - **Wetness:** Describes states from "Very wet, high moisture content (slurry-like)" to "Very dry and powdery" and "Dry."
 - **Size:** Describes textures from "Fine to coarse sand" to "Very fine-grained (appears like silt or very fine sand)" and "Predominantly fine-grained, appearing very clayey."

	CNN	LLM
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Accuracy	~70-80%	~15%
Pros	• Direct Image Feature Learning: CNNs are inherently designed for image processing. They excel at automatically learning hierarchical visual features (edges, textures, shapes) directly from raw pixel data, which is crucial for image-based classification tasks like soil identification. • High Classification Accuracy (for distinct classes): As seen in the t-SNE plot, the CNN demonstrated excellent separability and thus likely high accuracy for visually distinct soil types (for OA and OBOA classes). • Scalability: Once trained, CNNs can process new images very quickly for classification. • Robust to Variations: With sufficient training data, CNNs can be robust to minor variations in lighting, angle, or minor occlusions in images.	• Rich, Human-Readable Descriptions: The most significant advantage observed is the LLM's ability to generate detailed, qualitative descriptions of soil properties (colour, wetness, size/texture). This provides invaluable context and interpretability for domain experts, going far beyond a simple class label. • Contextual Understanding: LLMs, especially when integrated via Langchain, can potentially leverage broader knowledge bases (e.g., soil science literature, geological reports) to inform their analysis and descriptions, leading to more nuanced outputs. • Flexible Output Format: LLMs can generate structured data (like the key-value pairs seen) or natural language explanations, making their output highly adaptable to various downstream applications. • Potential for Multi-Modal Input: If the LLM is multi-modal, it can directly process image inputs and generate both classifications and descriptions. If not, it can process text descriptions derived from image analysis.

Cons

- **Struggles with Visual Ambiguity:**

The t-SNE plot for online datasets clearly showed significant overlap for several soil types (Clay, Laterite, Red, Mountain, Cinder). This indicates the CNN struggles to learn sufficiently discriminative features when visual differences are subtle or overlapping, leading to potential misclassifications.

- **Data Intensive:** Training high-performing CNNs requires very large, diverse, and accurately labelled image datasets.
- **Limited Descriptive Output:** The primary output is a classification label and a probability, not rich qualitative descriptions of the soil's properties (colour, wetness, texture).

- **Varying Confidence Levels:**

While the LLM consistently classified all samples as '2' (OBOA), its confidence scores varied widely, including very low confidence for some correct classifications. This suggests that even when correct, the model might be uncertain or struggling with certain visual nuances.

- **Classification Accuracy (across all classes):**

The provided output only showed classification for one soil type (OBOA). Its ability to accurately *differentiate* between multiple soil types (e.g., OA vs. OBOA) with high precision and recall across all classes is not fully demonstrated here.

- **Computational Cost:** Large LLMs can be computationally expensive for training and inference, requiring significant hardware resources.
- **Reliance on Input Quality:** The quality of the LLM's descriptions and classifications heavily depends on the quality and detail of the input it receives (whether it's raw image data for a multi-modal LLM, or pre-extracted features/descriptions for a text-based LLM).
- **Potential for "Hallucinations":** LLMs can sometimes generate plausible but incorrect information if

		not properly constrained or if the input is ambiguous.
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In summary, if the primary objective is to accurately assign a categorical label to an image based on its visual characteristics, **CNN is the superior choice due to its specialised architecture for image processing**, its proven ability to learn discriminative visual features, and its direct approach to classification. While LLMs offer unparalleled descriptive and interpretive capabilities, they are not inherently designed to be the optimal solution for the core task of visual classification when compared to a dedicated CNN. Hence why **CNN's accuracy is higher than LLM.**

The success of this project is largely dependent on the quality of the datasets used. An effective dataset should be diverse and large in quantity. Unfortunately, the CR111 dataset is limited in both size and variety, which leads to **results that should be interpreted with caution.**

To address these limitations, **it is crucial to obtain a higher-quality dataset, potentially using better hardware and cameras.** In terms of variety, the CR111 dataset falls short because Singaporean soil predominantly consists of Old Alluvium. A more suitable dataset would be from MRT Line 2, where the soil types along the tunnel stretch are distinct and varied.